Doping Stability Report -- Sb in LiCoO2 at 320 C

Generated: 2026-06-02 18:24   Host: LiCoO2   Dopant: Sb

Executive Summary

Sb preferentially occupies the Co layer. The formation energy difference between the two layers is +11.249 eV, strongly favouring the Co site. MD confirms this preference: on the Co site the dopant MSD plateau slope is -0.00178 A^2/ps (stable), compared to 0.06508 A^2/ps on the Li site. The Li-site E_f is approximate due to a charge surplus of +4 that cannot be modelled in CHGNet.

Site	E_f (eV)	Charge	Compensation	MSD pass	Coord pass	Verdict
Co	+6.402	+2	Yes	PASS	PASS	METASTABLE
Li	+17.651	+4	Yes	FAIL	FAIL	UNSTABLE (migration + structu

Sb at the Co layer -- METASTABLE

Charge situation

Sb replaces Co with a charge surplus of +2. CHGNet cannot model this compensation (would require Co oxidation or electron addition). The formation energy below is approximate.

Formation Energy (thermodynamic stability)

Formation energy E_f	**+6.4021 eV**	FAIL	*< 1.0 eV to pass*

E_f = +6.402 eV (approximate -- charge surplus not modelled in CHGNet). This value is large and positive, indicating that Sb incorporation is thermodynamically unfavorable under equilibrium conditions at the Co site.

Molecular Dynamics Stability (320 C, 25 ps)

MSD slope (late-time)	**-0.00178 A^2/ps**	PASS	*< 0.005 to pass (plateau = stable)*
Final MSD	**0.867 A^2**	----	
Max displacement	**1.991 A**	----	
Coordination (min/mean/max)	**6 / 6.0 / 7**	PASS	*relaxed = 6, tolerance +/-1*
Mean bond length (MD vs relaxed)	**2.003 A (relaxed: 1.9**	----	
Lattice volume change	**+0.00 %**	PASS	*< +/-5% to pass*

Late-time MSD slope = -0.00178 A^2/ps. The negative slope is a flat plateau -- essentially no migration detected. Final MSD = 0.867 A^2; maximum displacement from starting position = 1.99 A.

Coordination fluctuated between 6 and 7 (mean 6.0), within the +/-1 tolerance of the relaxed value (6). This is consistent with normal thermal vibration. Mean nearest-neighbour distance during MD = 2.003 A.

Volume change = +0.00%. The host lattice is mechanically intact.

VERDICT: METASTABLE

Sb sits stably on the Co site during MD at 320 C, but the formation energy exceeds the 1.0 eV equilibrium threshold. This configuration is metastable -- it may be achievable via non-equilibrium synthesis routes such as ion exchange or rapid quenching.

Sb at the Li layer -- UNSTABLE (migration + structural distortion)

Charge situation

Sb replaces Li with a charge surplus of +4. CHGNet cannot model this compensation (would require Co oxidation or electron addition). The formation energy below is approximate.

Formation Energy (thermodynamic stability)

Formation energy E_f	**+17.6508 eV**	FAIL	*< 1.0 eV to pass*

E_f = +17.651 eV (approximate -- charge surplus not modelled in CHGNet). This value is large and positive, indicating that Sb incorporation is thermodynamically unfavorable under equilibrium conditions at the Li site.

Molecular Dynamics Stability (320 C, 25 ps)

MSD slope (late-time)	**0.06508 A^2/ps**	FAIL	*< 0.005 to pass (plateau = stable)*
Final MSD	**0.240 A^2**	----	
Max displacement	**1.225 A**	----	
Coordination (min/mean/max)	**3 / 5.8 / 8**	FAIL	*relaxed = 6, tolerance +/-1*
Mean bond length (MD vs relaxed)	**2.196 A (relaxed: 2.2**	----	
Lattice volume change	**+0.00 %**	PASS	*< +/-5% to pass*

Late-time MSD slope = 0.06508 A^2/ps. The large slope indicates significant diffusion -- the dopant is migrating. Final MSD = 0.240 A^2; maximum displacement from starting position = 1.22 A.

Coordination fluctuated between 3 and 8 (mean 5.8), deviating by more than +/-1 from the relaxed value (6). This may indicate structural distortion, or that the bond-length cutoff is too close to the actual bond length -- check the coordination_cutoff_A setting. Mean nearest-neighbour distance during MD = 2.196 A.

Volume change = +0.00%. The host lattice is mechanically intact.

VERDICT: UNSTABLE (migration + structural distortion)

Verdict: UNSTABLE (migration + structural distortion). Refer to the individual criteria above.

Site Preference Comparison

The Co site is preferred by 11.249 eV over the Li site. Formation energy: +6.402 eV (Co) vs +17.651 eV (Li).

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Caveats and Limitations

? **CHGNet is charge-neutral.**

The potential does not model oxidation states, polaron formation, or electron/hole localisation. Formation energies for aliovalent dopants (charge mismatch != 0) are approximate even when structural compensation defects are included.

? **No Freysoldt/Kumagai correction.**

Electrostatic finite-size corrections for charged defects require the host dielectric tensor and the DFT electrostatic potential, neither of which is available from a neural-network potential. For publication-quality E_f of charged defects, follow up with DFT using doped or pymatgen-analysis-defects.

? **Metal-rich reference state.**

Chemical potentials are computed from elemental ground-state structures. Synthesis under oxide-rich or other conditions shifts all E_f values by additive constants; the relative ordering of sites is unaffected.

? **MD timescales are short (25 ps default).**

This is sufficient to assess local site stability and rapid migration, but cannot resolve slow diffusion mechanisms. A non-plateauing MSD confirms instability; a plateau does not guarantee stability on longer timescales.

? **Energy rankings are more trustworthy than absolute values.**

CHGNet is a universal MLIP trained on r2SCAN DFT. Formation-energy errors for transition-metal oxides are typically 50-150 meV/atom. Use relative orderings (which site is preferred?) as the primary result.